

ABSTRACT

Smart Greenhouse Automation System utilizes advanced technologies such as the Internet of Things (IoT), sensors, and Artificial Intelligence (AI) to automate and optimize the management of Greenhouse environments. The system integrates various components, including temperature, humidity, light, and soil moisture sensors, to monitor environmental conditions in real-time. These sensors are connected to a central control system that uses algorithms to make automated decisions, such as adjusting lighting, irrigation, and ventilation to create optimal growing conditions for plants. The use of IoT allows remote monitoring and control via mobile applications or web interfaces, enabling growers to oversee their greenhouse operations from anywhere. Machine Learning and AI models analyze data trends, helping predict crop needs and enhance growth efficiency. This system reduces human intervention, increases productivity, conserves resources like water and energy, and improves the quality and yield of crops. Ultimately, a Smart Greenhouse Automation System leverages technology to provide a sustainable and efficient solution for modern agriculture.

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ABBREVIATIONS

1. DHT - Digital Humidity and Temperature sensor
2. DS18B20 - Digital Temperature sensor
3. LDR - Light Dependent Resistor (Light sensor)
4. (SMS) - Soil Moisture Sensor
5. Wi-Fi - Wireless Fidelity (Wireless Communication)
6. MQTT - Message Queuing Telemetry Transport(IoT Protocol)
7. HTTP - Hypertext Transfer Protocol(Web Communication)

CHAPTER 1

INTRODUCTION

1. Overview

A Greenhouse is a building where plants are grown. Greenhouse protects crops from too much heat or cold, shield plants from dust storms and blizzards, and help to keep out pests. Light and temperature control allows Greenhouse to turn in into arable land, thereby improving food production in marginal environments. Because Greenhouse allow certain crops to be grown throughout the year, Greenhouses are increasingly important in the food supply of high latitude countries. Greenhouses are often used for growing flowers, vegetables, fruits and tobacco plants.

2. Problem Statement

- Prior Greenhouse systems were not using microcontroller. Means the systems were not automatic. User checks the conditions and takes proper action.
- After sometime the Greenhouse systems were made automatic means system will work for the user and takes proper action to maintain climatic conditions. But monitoring was done by live presence of user.
- After some time monitoring was made easy and user was able to monitor the green house from his home. Here the distance from the system was limited.
- Manufacturers use Bluetooth app or system specific screen having data of Greenhouse climatic conditions. Here also the monitoring becomes bounded.

3. Objectives

A Greenhouse is an important part of the agriculture and horticulture sectors in our country. It is used for growing plants faster at any season whether the climate is according to plant or not. Automatic monitoring and controlling of the climate parameters will directly or indirectly govern the plant growth and hence their production.

Project Background

The closed environment of the Greenhouse has its own unique requirements, compared with outdoor production. Pests and diseases, and extremes of the heat and humidity, have to be controlled, and irrigation is necessary to provide water. Significant inputs of the heat and light may be required, particularly with the winter production of warm weather vegetables. Because the temperature and humidity of greenhouses must be constantly monitored to ensure optimal conditions, a wireless sensor network can be used to gather data remotely. The data is transmitted to a control location and used to control heating, cooling, and irrigation systems. Thus, a wireless sensor network consists of small size wireless sensor nodes equipped with radio and one or several sensors in an attractive and cost efficient option to build the required measurement system. Wireless technologies have been rapidly developed during recent years.

Enhancements and Future Scope

While the current project is a prototype, there is significant potential for future enhancements. Improving accuracy in diverse lighting and background conditions. Enabling multi-player gameplay for increased interaction. Using advanced machine learning models for more accurate gesture recognition.

Scope of the Project

CHALLENGES:

Maintaining the temperature due to change in climate the regular temperature may vary again and again plants growing inside the Greenhouse temperature need to be maintained. Requires constant monitoring as all the values are updated to cloud there should be proper monitoring to all sensor and hardware.

MOTIVATION:

The purpose of a green house is to shield crops from excess cold, heat and unwanted pests. A Greenhouse makes it possible to grow certain types of crops year round. Within the framework of sustainable development the energy supply system is a crucial topic. Spurred on by the Kyoto Protocol the attention for the definition of energy saving programs. Promotion of environment-friendly technologies as renewable energy is increasing.

CHAPTER 2

LITERATURE SURVEY

1 Paul D. Rose ro Montalvo And et al.

Journal: IEEE Access (DOI : 10.1109/ACCESS.202 3.3283986 Date : 8 June 2023).

To efficiently manage agricultural fields and greenhouses today, farmers have to apply technologies in line with Industry 4.0, such as: robots, Internet of Things devices, machine learning applications, and so on. An unsupervised learning algorithm is implemented to cluster the optimal, standard, and deficient sectors of a greenhouse to determine inappropriate growth patterns in crops.

2 Muh amm ad Shoa ib Faro oq , And et al.

Journal: IEEE Access (DOI : 10.1109/ACCESS.202 2.3166634 Date : 11 April 2022) .

A rigorous discussion on greenhouse farming techniques, IoT-based greenhouse categories, network technologies , the success stories and statistical analysis of some agricultural countries have been presented to standardize IoT-based greenhouse farming.

3 Jun Jiang , Meh rdad Moal lem

Journal: IEEE Access (DOI : 10.1109/IECON43393 .2020.9254993 Date : 20 June 2021).

To develop an intelligent control system for mixing colour ratios using LED lights in a greenhouse environment. To this end, different components of an experimental testbed is presented for achieving the desired light requirements for plant growth in a greenhouse environment. The proposed testbed provides a easy-to-use plant growth system with IoT-enabled control and monitoring features. To testify the features mentioned above, a feedback lighting control method to achieve a desired photosynthetic photon flux density (PPFD) set point is implemented.

4 Rand a Osa ma , Nour El Hud a and et al.

Journal: IEEE Access (DOI : 10.1109/NILES50944 .2020.9257974 Date : 16 June 2021)

The proposed system is used to speed up the plant growth and detect the plant' s diseases. We used tomatoes to test our proposed system. The detected diseases are early blight, late blight, leaf mold, spider mites, target spot, mosaic virus, septoria, bacterial spot, and yellow leaf curl virus.

5 Prad yum na K. Trip athy , and et al

Journal : IEEE Access (DOI : 10.1109/MCE.2021.3 055930 Date : February 2021)

This work also accounts for the different challenges of greenhouse rose farming and highlights a new IoT-based solution, which is smart and sustainable.

6 Radh a N ,Swa thika R Journal: IEEE Access

The preliminary goal is to identify the type of plant disease. The proposed plant model is used to detect the diseased portions of a plant and also monitors the parameters related to crop production.

7 Ding zhu Xue 1, and Wei Hua ng Journal: IEEE Xplore

Introduced the classification of positioning algorithm, and then analyzed the DV-HOP positioning algorithm. Aiming at the problem of low positioning accuracy and large error of DV-HOP algorithm, an improved method of DV-HOP algorithm based on average HOP distance was proposed to make positioning more precise.

8 Sand unik a Fern ando Ranu sha Neth mi , and et al Journal: IEEE Xplore

This research is focused on diseases that cause yellowish the tomato plants inside greenhouses. There are several reasons for the yellowness in the tomato plant leaves. The system identifies leaf mold, tomato yellow leaf curl, diseases, and healthy leaves.

9 Ahm ad F. Suba hi And Khei r Eddi ne Boua zza Journal: IEEE Access

In Deep Neural network makes accurate predictions by extracting useful features of the input image. Activation maps or feature maps are obtained by applying filters to the input image by each layer of the CNN model.

CHAPTER 3

METHODOLOGY

1. System Requirements

1. Functional Requirements:

- System should automatically read temperature
- System should automatically read Humidity
- System should automatically read moisture level and lighting condition

3.1.2 Non Functional Requirements:

- Usability: Easy Interface for user
- Reliability: Theft Avoidance
- Performance: Should not take excessive time in taking measure

3.2 Hardware Requirements

- Microcontroller
- Moisture Sensor
- DHT11 Sensor
- Power Supply
- DC Fan
- LCD
- Relay

3.3 Software Requirements

- Embedded C
- ESP 32

CHAPTER 4:

DESIGN AND DEVELOPMENT

1. System Architecture

System Architecture design-identifies the overall hypermedia structure for the Application. Architecture design is tied to the goals establish for a WebApp, the content to be presented, the users who will visit, and the navigation philosophy that has been established. Content architecture, focuses on the manner in which content objects and structured for presentation and navigation. WebApp architecture, addresses the manner in which the application is structure to manage user interaction, handle internal processing tasks, effect navigation, and present content. Application architecture is defined within the context of the development environment in which the application is to be implemented.

2. ESP32 bit Micro-Controller

ESP32 is a microcontroller board. ESP32 devkit has 36 pins and 18 on each side of the board as shown in the picture above. It has 34 GPIO pins and each pin has multiple functionalities which can be configured using specific registers. There are many types of GPIOs available like digital input, digital output, analog input, and analog output, capacitive touch, UART communication and USB cable.

3. LCD: Liquid Crystal Display:

This is an LCD Display designed for E-blocks. It is a 16 character, 2-line alphanumeric LCD display connected to a single 9-way D-type connector. This allows the device to be connected to most E-Block I/O ports. The LCD display requires data in a serial format, which is detailed in the user guide below. The display also requires a 5V power supply. Please take care not to exceed 5V, as this will cause damage to the device. The 5V is best generated from the E-blocks Multi programmer or a 5V fixed regulated power supply. The potentiometer RV1 is a contrast control that should be used to adjust the contrast of the display for the environment it is being used in. The LCD is a 16 character x 2 lines module. Internally it is 40 characters x 2 lines. Line 1 rangis from H' 80' to H' 8F' and Line 2 ranges from H' 0C' to H' CF' . The LCD Module uses a Samsung KS0066U controller, which is similar to the Hitachi HD4478.

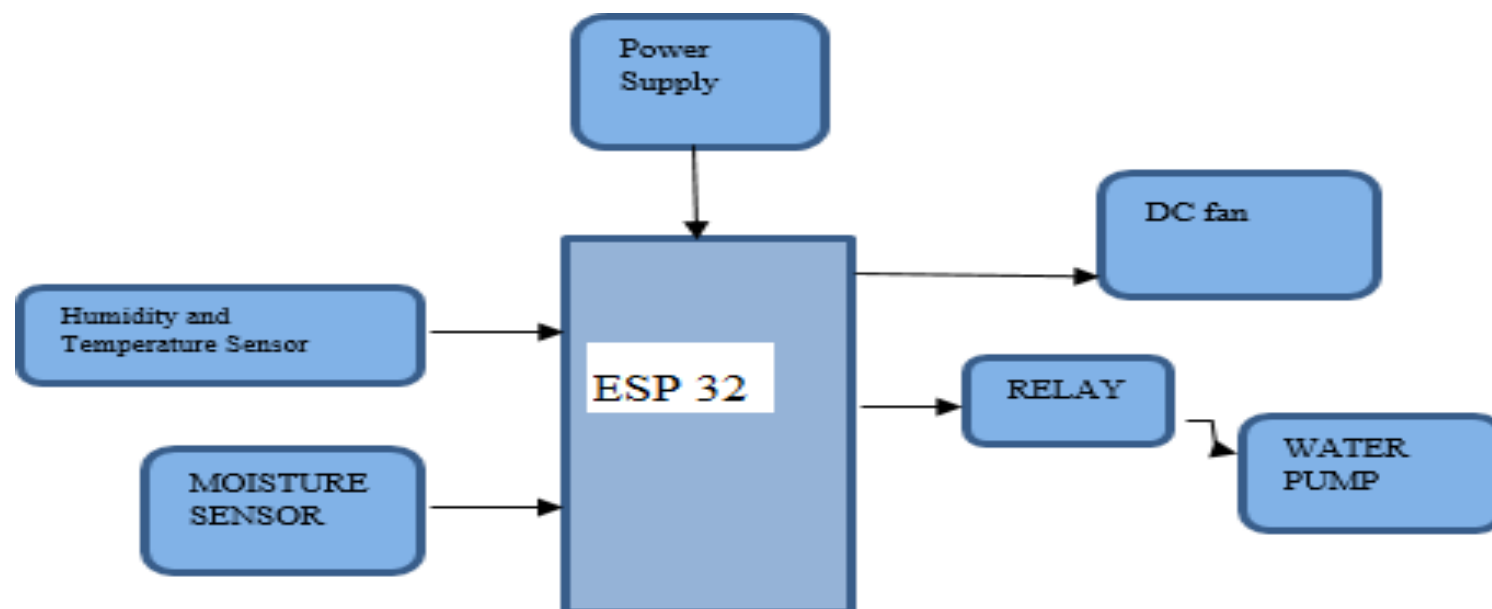


Figure.4.1: System block diagram of crop protection and rainwater harvesting

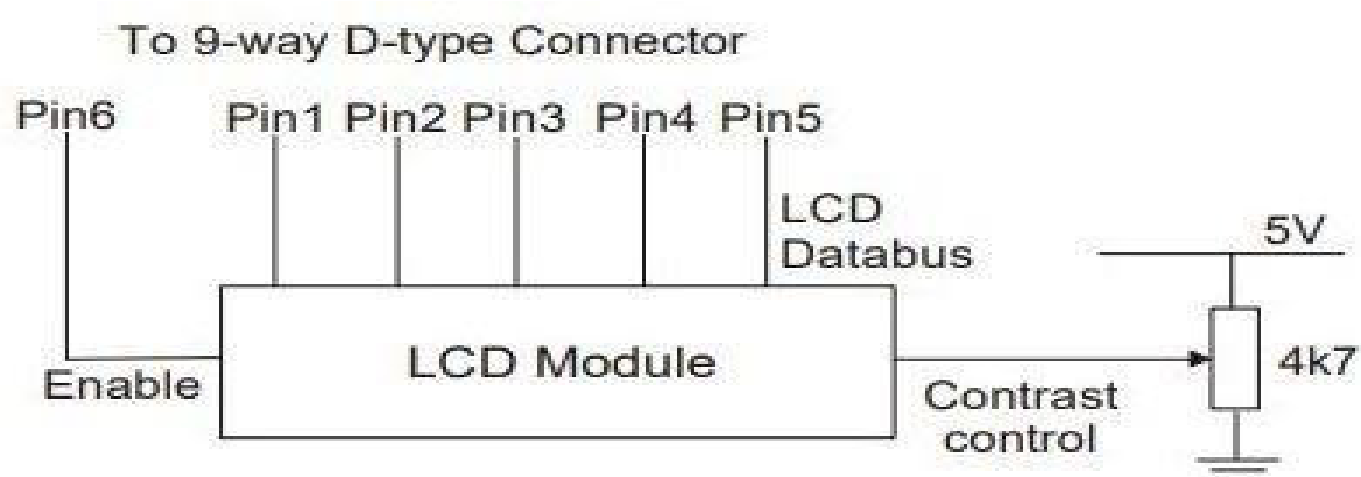


Figure.4.2: LCD block diagram

To send a command to the LCD, data must be sent in two steps, the MSB followed by the LSB. As each byte is sent to the LCD, B5 must be go high then low, for the LCD to acknowledge the byte. After the second byte has been acknowledged the LCD executes the command. The PIC micro board must wait for at least the length of the execution time for that command, before the next command can be sent

Features :

- E-blocks compatible
- Low cost
- Compatible with most I/O ports in the E-Block range
- Easy to develop programming code using flowcode icons

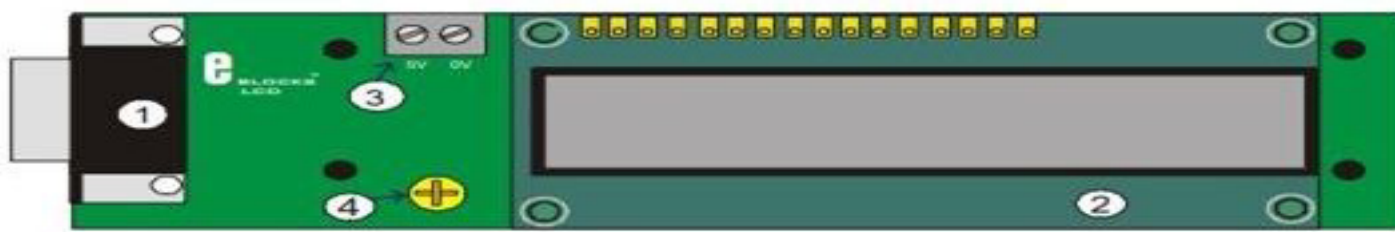


Figure.4.3: LCD pin diagram

- **Threshold:** The mask is threshold to create a binary image, where the skin areas are white, and the non-skin areas are black. This simplifies the hand region detection.
- **Morphological Operations:** Dilation and Gaussian blur are applied to remove small noise and fill gaps in the mask, which enhances the accuracy of hand detection and ensures the contours are smooth and complete.

4.4 Moisture sensor:

Soil moisture sensor measures the water content in soil. It uses the property of the electrical resistance of the soil. Here, it is used to sense the moisture in field and transfer it to microcontroller in order to take controlling action of switching water pump ON/OFF.

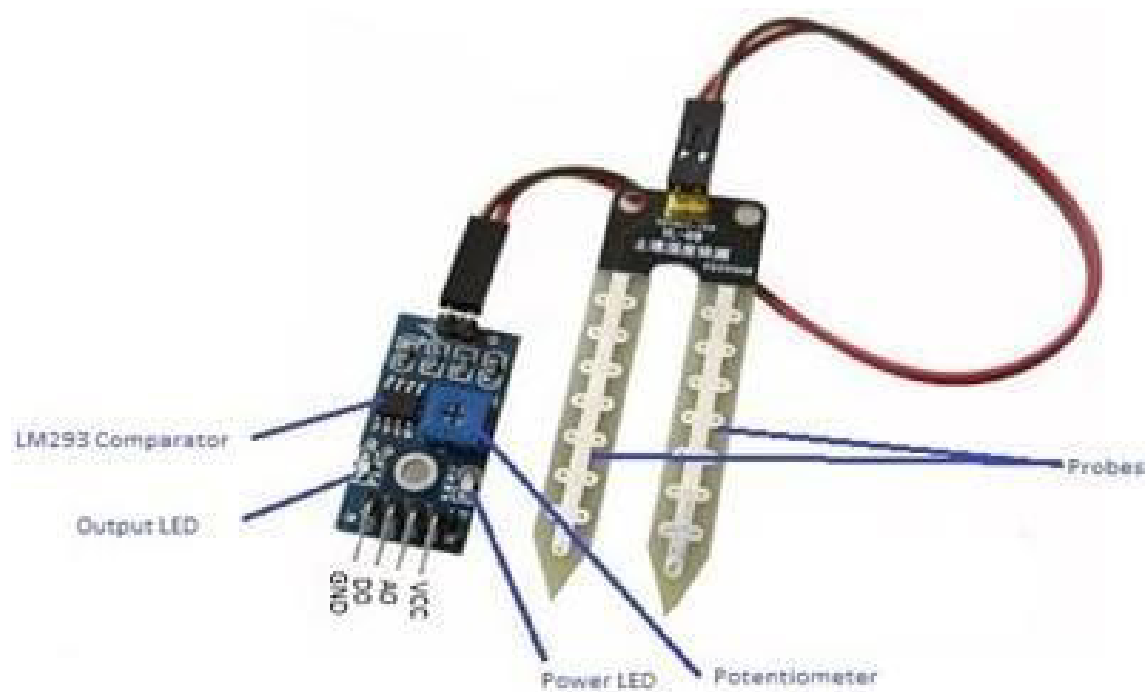


Figure.4.4: Moisture Sensor

Soil moisture sensor

Soil Moisture Sensor measure the volumetric water content in soil Since the direct gravimetric measurement

of free soil moisture requires removing, drying, and weighing of a sample, soil moisture sensors measure the volumetric water content indirectly by using some other property of the soil, such as electrical resistance, dielectric constant, or interaction with neutrons, as a proxy for the moisture content.

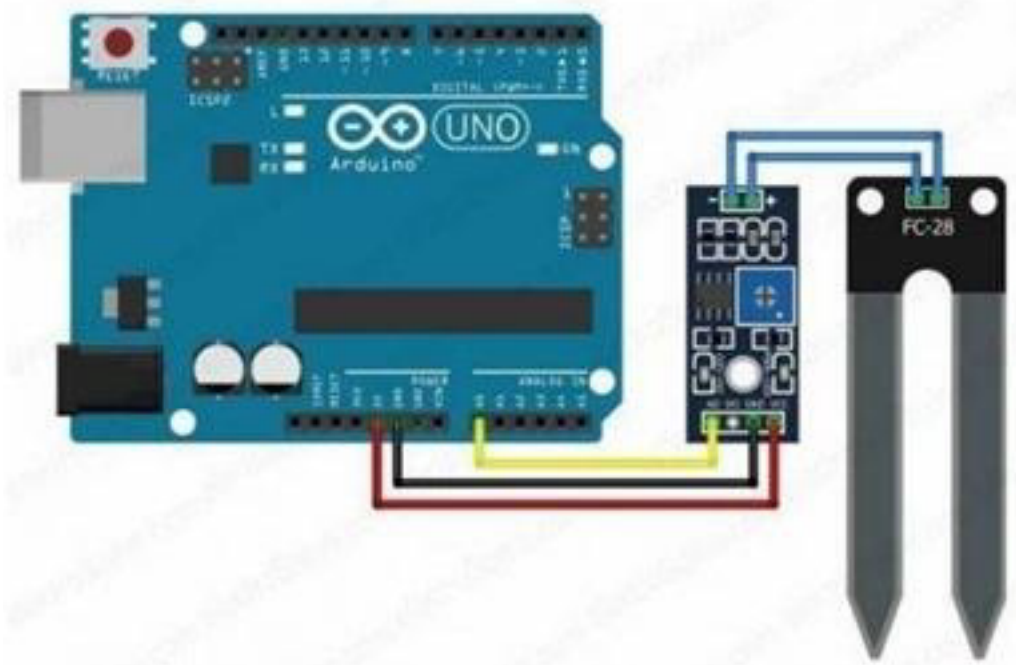


Fig 4.5 Soil moisture sensor with ESP 32

The sensor contains a fork-shaped probe with two exposed conductors that goes into the soil or anywhere else where the water content is to be measured. The module produces an output voltage according to the resistance of the probe and is made available at an Analog Output (AO) pin. The fork-shaped probe with two exposed conductors, acts as a variable resistor (just like a potentiometer) whose resistance varies according to the water content in the soil. This resistance is inversely proportional to the soil moisture. The more water in the soil means better conductivity and will result in a lower resistance. The less water in the soil means poor conductivity and will result in a higher resistance. The sensor produces an output voltage according to the resistance. The sensor produces an output voltage according to the resistance, which by measuring we can determine the moisture level.

4.5 DHT-11 Sensor

DHT11 Temperature & Humidity Sensor features a temperature & humidity sensor complex with a calibrated digital signal output. By using the exclusive digital-signal-acquisition technique and temperature & humidity sensing technology, it ensures high reliability and excellent long-term stability. This sensor includes a resistive- type humidity measurement component and an NTC temperature measurement component, connects to a high performance 8-bit microcontroller, offering excellent quality, fast response, anti-interference ability and cost-effectiveness.

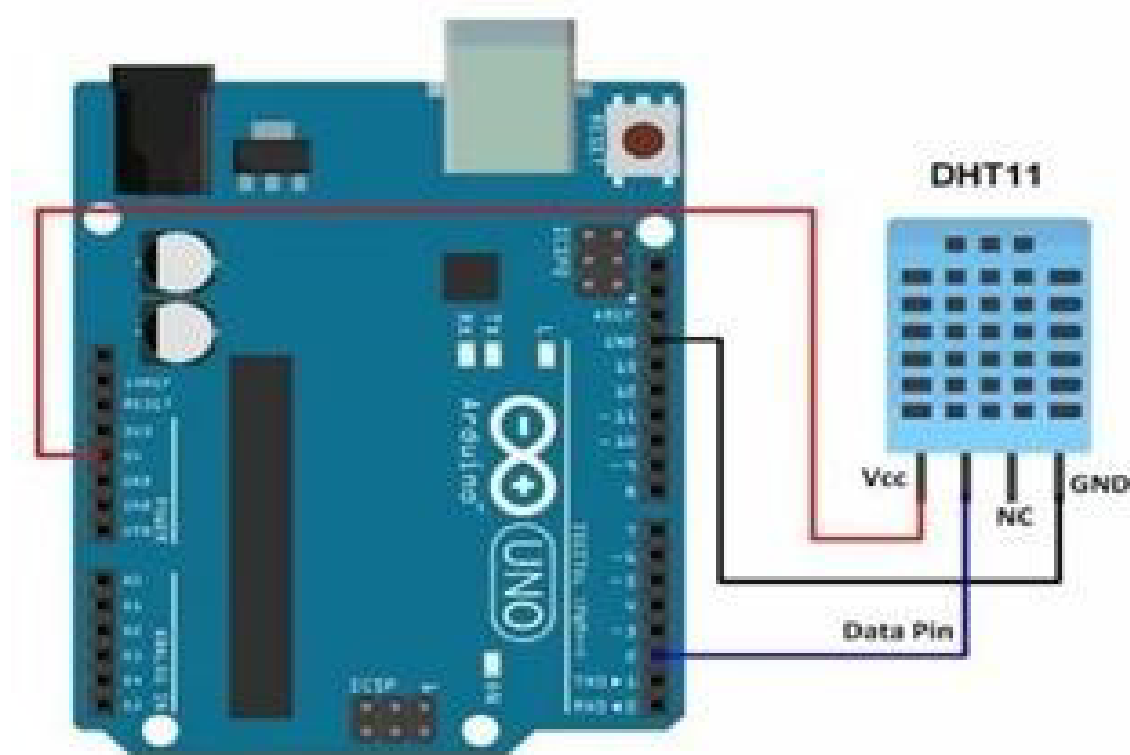


Fig 4.6 DHT-11 sensor with ESP 32

Specifications

- Humidity Range: 20-90%RH.
- Humidity Accuracy: $\pm 5\%$ RH.
- Temperature Range: 0-50° C
- Temperature Accuracy: $\pm 2\%$ ° C
- Operating Voltage: 3V to 5.5V

LCD display

LCD (Liquid Crystal Display) is a type of flat panel display which uses liquid crystals in its primary form of operation.

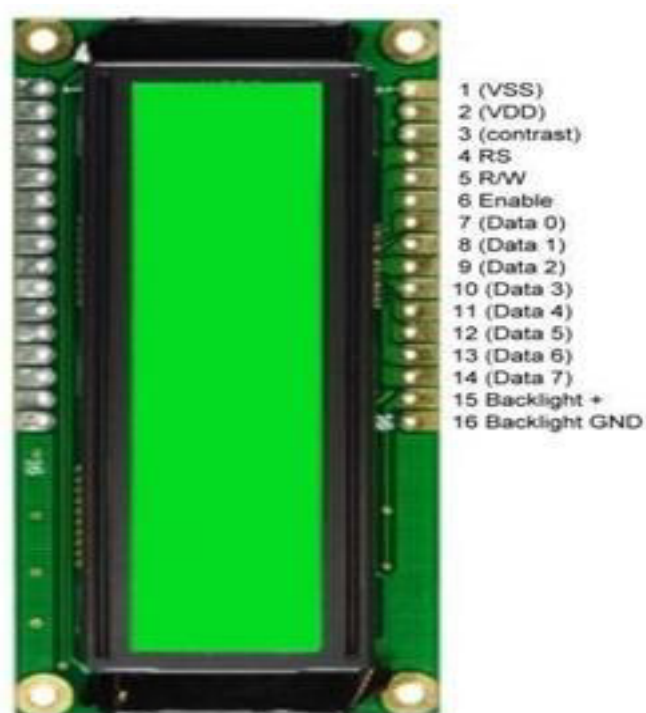


Fig 4.7 LCD socket pin connections

Pin mapping and functions

Normally to interface LCD with the microcontroller requires 3 control signals and 8 data lines. This is known as 8bit interfacing mode which requires total 11 I/O lines. To save number of I/O s required for LCD interfacing we can use 3 control signals with 4 data lines. This is known as 4bit interfacing mode and it requires 7 I/O lines. We are using 4bit interfacing mode to reduce number of I/O lines. In this mode higher nibble and lower nibble of commands/data set needs to be sent separately. Figure shows LCD interfacing in 4bit mode. The three control lines are referred to as EN, RS, and RW. The EN line is called “ Enable” and it is connected to PORT2’ s 5th pin (P2.4). The control line issued to tell the LCD that microcontroller has sent data to it or microcontroller is ready to receive data from LCD. This is indicated by a high-to-low transition on this line. To send data to the LCD, program should make sure that this line is low (0) and then set the other two control lines as required and put data on the data bus. When this is done, make EN high (1) and wait for the minimum amount of time as specified by the LCD datasheet and end by bringing it to low (0) again.

Pin No	Function	Name
1	Ground (0V)	Ground
2	Supply voltage; 5V (4.7V – 5.3V)	V _{CC}
3	Contrast adjustment; through a variable resistor	V _{EE}
4	Selects command register when low; and data register when high	Register Select
5	Low to write to the register; High to read from the register	Read/write
6	Sends data to data pins when a high to low pulse is given	Enable
7	8-bit data pins	DB0
8		DB1
9		DB2
10		DB3
11		DB4
12		DB5
13		DB6
14		DB7
15	Backlight V _{CC} (5V)	Led+
16	Backlight Ground (0V)	Led-

Table 4.8 LCD socket pins description

The RW line is the 'Read/write' control line and it is connected to PORT 2's 6th pin (P2.5). When it is low(0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading from) the LCD. The data bus is bidirectional, 4bit wide and is connected to PORT2(P2⁰ to P2³) of the microcontroller, The MSB bit (DB7) of data bus is also used as a busy flag. When the Busy flag is 1, the LCD is in internal operation mode, and the next instruction will not be accepted. When RS=0 and R/W=1, the Busy flag is output on DB7. The next instruction must be written after ensuring that the busy flag is 0.

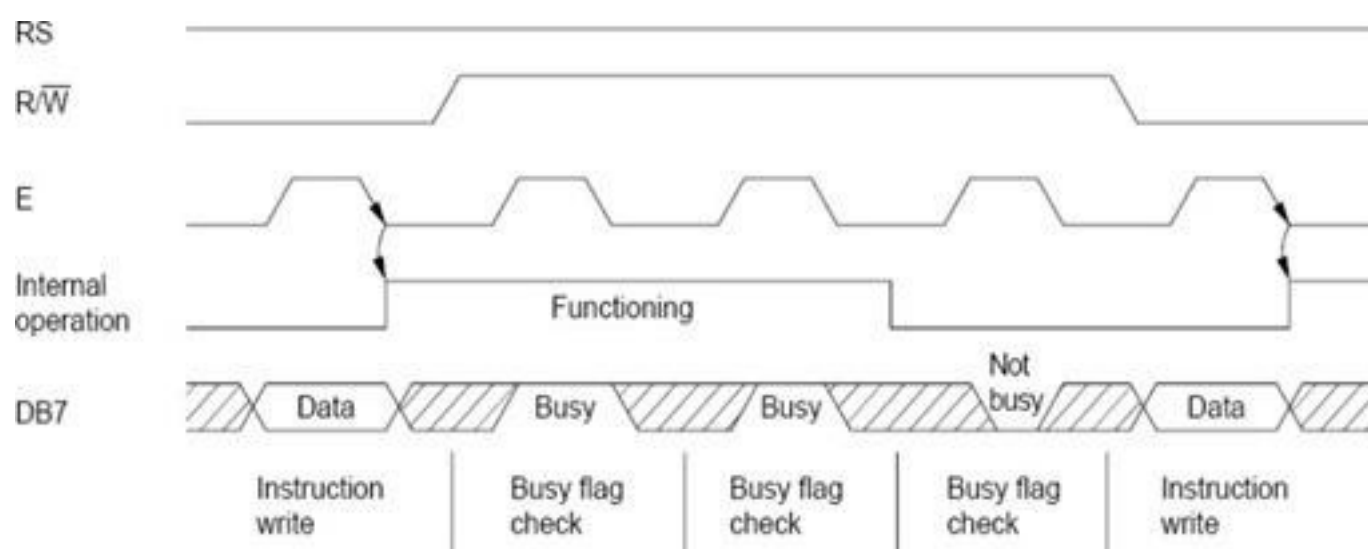


Fig 4.9 LCD timing diagram

4.6 ESP 8266:

ESP8266 offers a complete and self-contained Wi-Fi networking solution, allowing it to either host the application or to offload all Wi-Fi networking functions from another application processor. When ESP8266 hosts the application, and when it is the only application processor in the device, it is able to boot up directly from an external flash. It has integrated cache to improve the performance of the system in such applications, and to minimize the memory requirements. Alternately, serving as a Wi-Fi adapter, wireless internet access can be added to any microcontroller-based design with simple connectivity through UART interface or the CPU AHB bridge interface. ESP8266 on-board processing and storage capabilities allow it to be integrated with the sensors and other application specific devices through its GPIOs with minimal development up-front and minimal loading during runtime. With its high degree of on-chip integration, which includes the antenna switch balun, power management converters, it requires minimal external circuitry, and the entire solution, including front-end module, is designed to occupy minimal PCB area.

4.7 Water Pump:

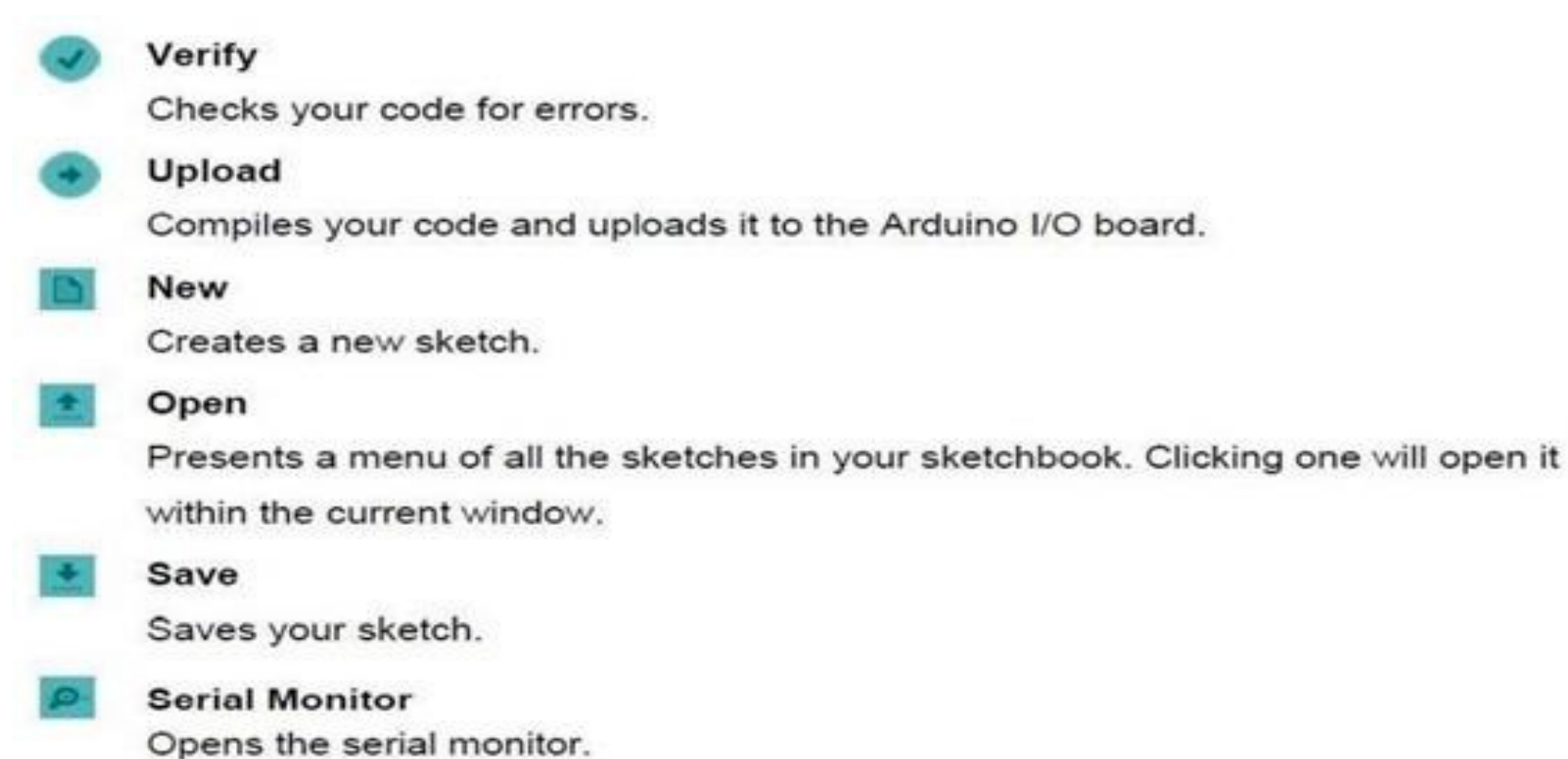
Micro DC 3-6V Micro Submersible Pump Mini water pump For Fountain Garden Mini water circulation System DIY project. This is a low cost, small size Submersible Pump Motor which can be operated from a 3~6V power supply. It can take up to 120 liters per hour with very low current consumption of 220mA. Just connect tube pipe to the motor outlet, submerge it in water and power it. Make sure that the water level is always higher than the motor. Dry run may damage the motor due to heating and it will also produce noise.

Specifications :-

- Operating Voltage : 3 ~ 6V
- Operating Current : 130 ~ 220mA
- Flow Rate : 80 ~ 120 L/H
- Maximum Lift : 40 ~ 110 mm
- Continuous Working Life : 500 hours
- Driving Mode : DC, Magnetic Driving
- Material : Engineering Plastic
- Outlet Outside Diameter : 7.5 mm
- Outlet Inside Diameter : 5 mm

ESP 32 IDE software

The ESP 32 IDE is a cross-platform application written in Java, and is derived from the IDE for the Processing programming language and the Wiring project. It is designed to introduce programming to artists and other newcomers unfamiliar with software development.



```
lcd.print(humidity); lcd.setCursor(0, 1); lcd.print("TEMP:"); lcd.setCursor(6, 1);  
lcd.print(temperature); delay(1000);  
if (temperature > 35) {  
    lcd.clear(); lcd.setCursor(0, 0); lcd.print("High Temp"); digitalWrite(fan, HIGH); delay(1000);  
} else {  
    digitalWrite(fan, LOW); delay(1000);  
}  
}
```

UNDERSTANDING THE CODE

1. Initialization (setup)

- Sets up the LCD, serial communication, and pin modes for the components:
- LCD: Displays information.
- Pins: Configured for sensors (LDR, moisture sensor, DHT11) and actuators (pump, fan, light).
- Displays a "Green House Monitoring" message.

2. Main Loop (loop)

Repeatedly checks and responds to:

- Temperature and Humidity (DHT_CHECK)
- Soil Moisture (MOISTURE)
- Light Intensity (LDR_CHECK)

3. Individual Functionality

a. DHT_CHECK

- Reads temperature and humidity from the DHT11 sensor.
- Displays the values on the LCD.
- Turns the fan *ON* if the temperature exceeds 35° C; otherwise, keeps it OFF.

b. MOISTURE

- Reads soil moisture from an analog sensor.
- Displays moisture value on the LCD.
- Turns the pump *ON* if moisture is low; otherwise, keeps it OFF.

c. LDR_CHECK

- Checks light levels using the LDR.
- Turns the light *ON* if light is low; otherwise, keeps it *OFF*.

4. Key Concepts

- ▶ **Sensors:** Detect environmental parameters.
 - DHT11 → Temperature and humidity.
 - LDR → Light levels.
 - Moisture sensor → Soil moisture.
- ▶ **Actuators:** Respond to sensor readings.
 - Fan → Cool the environment.
 - Pump → Water plants.
 - Light → Provide illumination.

How It Works Together

1. Monitor Conditions: Continuously checks temperature, humidity, light, and soil moisture.
2. Control Outputs: Activates or deactivates the fan, pump, and light based on conditions.
3. User Feedback: Provides real-time updates on the LCD and Serial Monitor.

Shortcut Understanding

- Inputs: Sensors provide data (DHT11, LDR, Moisture sensor).
- Logic: Conditional checks for thresholds (e.g., temperature > 35° C, low light, low moisture).
- Outputs: Actuators (fan, pump, light) are controlled accordingly.
- User Interface: Displays status on the LCD and logs data on the Serial Monitor.

CHAPTER 5:

RESULT AND ANALYSIS

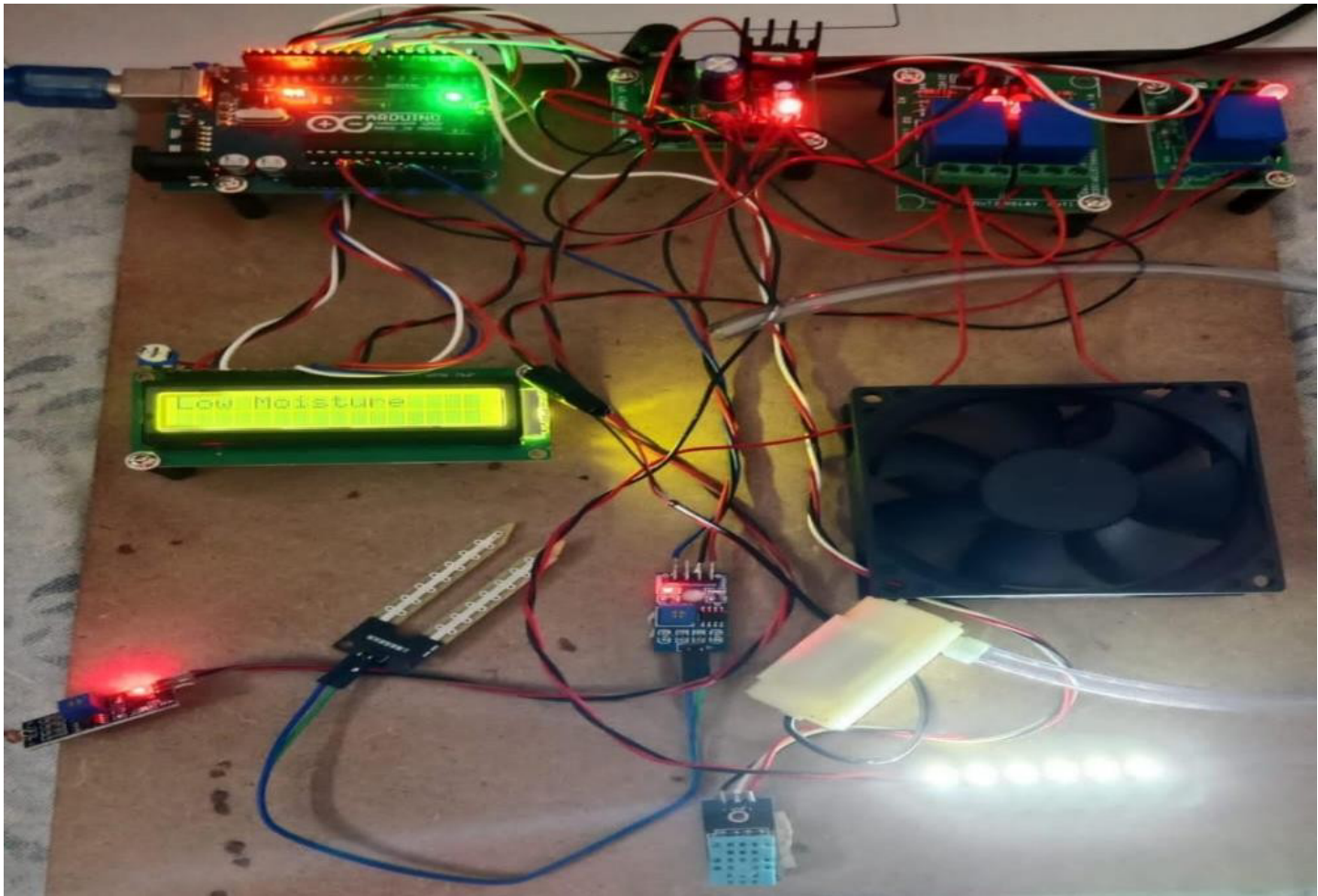


Figure 5.1:Smart Greenhouse Automation System

1. Project Overview

The greenhouse automation system monitors environmental parameters (temperature, humidity, light intensity, and soil moisture) and automates control mechanisms (fan, pump, and artificial light). The goal is to maintain optimal conditions for plant growth with minimal human intervention.

2. Observations and Results

a. Temperature and Humidity Monitoring

- Input: DHT11 sensor measures temperature and humidity.
- Output: The fan turns ON when temperature exceeds 35° C, providing cooling.
- Observation
- The fan activation efficiently reduced high temperatures within minutes.
- Humidity levels were displayed but not actively controlled in this iteration.

b. Soil Moisture Monitoring

- Input: Analog moisture sensor measures soil moisture levels.
- Output: The water pump activates when moisture falls below a threshold (e.g., 600).

- Observation
- The pump was triggered accurately during dry soil conditions.
- Prevented overwatering by turning off when soil moisture reached acceptable levels.

c. Light Intensity Monitoring

- Input: LDR sensor checks ambient light levels.
- Output: Artificial light activates in low-light conditions.
- Observation:
- The system detected low light correctly, ensuring plants received sufficient light.
- Light deactivation occurred when ambient conditions improved.

d. User Feedback

- LCD Display: Real-time updates of parameters (temperature, humidity, moisture, light conditions).
- Serial Monitor: Logged data for debugging and analysis.

3. System Performance Analysis

a. Accuracy and Reliability

- Sensors: Reliable readings for temperature, humidity, soil moisture, and light levels.
- Response: Timely activation/deactivation of actuators.

Limitations:

- DHT11 sensors may lag slightly in detecting rapid temperature changes.
- LDR is influenced by external light sources, which may need calibration.

a. Efficiency

- Automated controls reduce manual intervention and resource wastage (water, energy).
- System response was consistent in maintaining optimal greenhouse conditions.

b. Scalability

- Additional features (e.g., automatic humidity control, advanced sensors) can be integrated.
- Data logging for long-term analysis could enhance decision-making.

c. Power Consumption

- Minimal power usage by sensors and actuators ensures cost-effective operation.
- Incorporating solar power could further improve sustainability.

CHAPTER 6:

CONCLUSION

The Greenhouse Automation System effectively demonstrates the use of sensors and actuators to monitor and control environmental conditions such as temperature, humidity, soil moisture, and light intensity. By automating tasks like watering, cooling, and artificial lighting, the system reduces manual labor and optimizes resource usage, ensuring plants grow in a controlled and favorable environment. The project successfully achieves its objectives, providing a cost-effective, energy-efficient, and user-friendly solution for small-scale greenhouses.

The greenhouse automation system has significant potential for further development and scalability. Some potential future improvements include:

1. IoT Integration

Enable remote monitoring and control using mobile apps or web interfaces. Incorporate cloud-based data storage for real-time updates and historical data analysis.

2. Advanced Sensor Technology

Replace basic sensors like the DHT11 with more accurate and reliable sensors (e.g., DHT22, BMP280, or capacitive soil moisture sensors). Add additional sensors for CO₂ levels, pH, and nutrient monitoring.

3. Energy Optimization

Integrate renewable energy sources such as solar panels to make the system self-sustaining. Implement power-saving algorithms to minimize energy consumption.

4. Automated Fertilization

Add a mechanism for dispensing nutrients based on soil condition monitoring.

5. Predictive Maintenance

Use data analytics to predict equipment failures and ensure smooth operation.

6. Scalability for Larger Applications

Adapt the system for larger commercial greenhouses by using wireless sensor networks (WSNs) and distributed control systems.

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